

Question

Sodium salt Na_2X contains a unique cyclic anion, in which each atom in the anion has the same chemical environment. Sodium salt Na_2X is often used to test metal cations such as Ba^{2+} and Pb^{2+} , and even when in contact with the surface of certain insoluble lead salts (such as PbS), it can seize Pb^{2+} from them and form a characteristic purplish-red substance. Na_2X is unstable to both light and heat and is very sensitive to oxidizing environments. Under acidic conditions, I_2 reacts with Na_2X in equal proportions, yielding only NaI and gas **M**; KMnO_4 reacts with Na_2X , yielding gas **N**. If the gas **M** obtained from the reaction of 1.498 g Na_2X with I_2 is passed through sufficient I_2O_5 , I_2O_5 decreases in mass by 0.672 g. A weakly acidic PbAc_2 solution is divided into two portions, one with Na_2X , forming a basic salt precipitate **A** containing one water of crystallization (in which the stoichiometric ratio of each ion contained is 2 : 1 : 2), precipitate **A** is scarlet; the other

is added with sufficient Na_2CO_3 to obtain precipitate **B**, heating precipitate **B** in aqueous solution yields precipitate $\text{Pb}_3(\text{OH})_2(\text{CO}_3)_2$ with the same elemental composition as **A**.

Please select the option below that best fits the meaning of the question:

- A.** All other options are incorrect
- B.** **A** The chemical formula after removing water of crystallization contains 16 atoms.
- C.** X^{2-} ion does not possess aromaticity.
- D.** The chemical equation for the reaction of Na_2X with I_2 , when balanced with the simplest integer ratio, has a sum of product coefficients equal to 6.
- E.** The ion equation for the reaction of Na_2X with acidic KMnO_4 , when balanced to the simplest integer ratio, has a sum of product coefficients equal to 60.

F. Ions Y^{2-} that are structurally similar to X^{2-} and have a relative molecular mass 28.01 smaller exhibit extremely low stability.

G. Multiple options are correct among options B-F.

Answer

Correct Answer: **A**

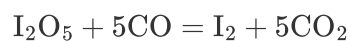
Detailed Explanation

The starting point of this question is to use I_2O_5 to test the gas **M**. According to chemical knowledge, the gas **M** is CO , and the reaction is quantitative:

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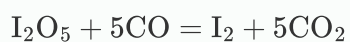
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M is CO



CHECKPOINT

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According to the mass given in the question, the amount of substance of CO produced is

$$0.672/16 \times 5 = 0.042 \text{ mol}$$

I_2 reacts with Na_2X in equal proportions to obtain NaI and CO , indicating that the cyclic anion X^{2-} is composed only of carbon and oxygen elements;

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Cyclic anion $\mathbf{X}^{2-}$ is composed only of carbon and oxygen elements;

1.498 g $\text{Na}_2\mathbf{X}$ can produce 0.042 mol of CO, that is, the content of sodium element in $\text{Na}_2\mathbf{X}$ is $1.498 - 0.042 \times 28.01 = 0.322 \text{ g} = 0.014 \text{ mol}$, thus the relative molecular mass of $\mathbf{X}^{2-}$ is calculated as $(1.498 - 0.322)/(0.014/2) = 168$, which is six CO, so the chemical formula of $\mathbf{X}^{2-}$ is

$\text{C}_6\text{O}_6^{2-}$.

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The relative molecular mass of $\mathbf{X}^{2-}$ is 168

CHECKPOINT

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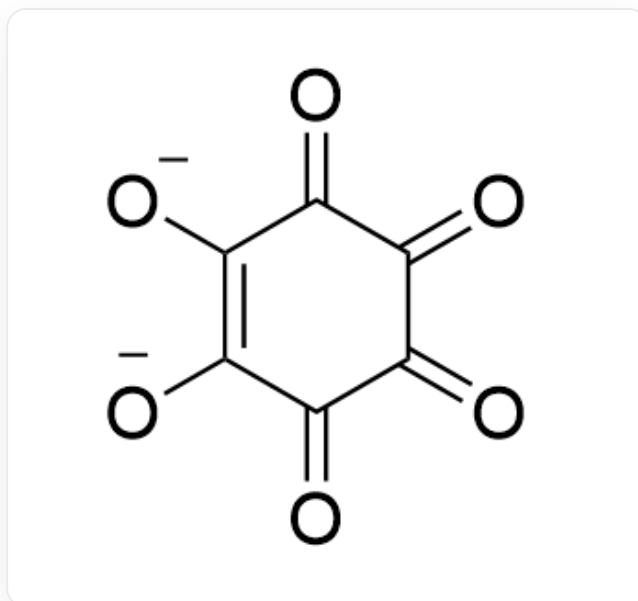
The chemical formula of $\mathbf{X}^{2-}$ is $\text{C}_6\text{O}_6^{2-}$

Since $\mathbf{X}^{2-}$ is a cyclic anion, it can be deduced that it contains a six-membered carbon ring, composed of six carbonyl groups; the structure is $\text{O}=\text{C}(\text{C}([\text{O}-])=\text{C}([\text{O}-])\text{C}(\text{C}=\text{O})=\text{O})\text{C}=\text{O}$, the six carbons/oxygens have the same chemical environment due to tautomerism, which is consistent with the description of the question.

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The structure of $\mathbf{X}^{2-}$ is $\text{O}=\text{C}(\text{C}([\text{O}-])=\text{C}([\text{O}-])\text{C}(\text{C}=\text{O})=\text{O})\text{C}=\text{O}$



The structure of X^{2-} : O=C(C([O-])=C([O-])C(C1=O)=O)C1=O

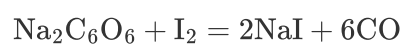
In the structure of O=C(C([O-])=C([O-])C(C1=O)=O)C1=O, there is a pair of electrons delocalized on the six-membered ring, which conforms to the $4n + 2$ rule ($n = 0$) and has certain aromaticity, so option C is incorrect.

CHECKPOINT

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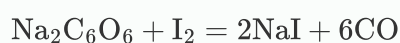
X^{2-} has a pair of electrons delocalized on the six-membered ring, and has certain aromaticity

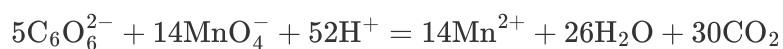
Therefore, it is easy to write the reaction equations of $\text{C}_6\text{O}_6^{2-}$ with iodine and acidic potassium permanganate:



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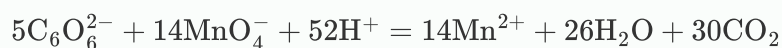
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According to the equation, options D and E are both incorrect.

The ion Y^{2-} with a relative molecular mass of 28.01 less than that of $\text{C}_6\text{O}_6^{2-}$ is obviously $\text{C}_5\text{O}_5^{2-}$, which becomes a five-membered ring $[\text{O-}]\text{C}(\text{C}(\text{C}1=\text{O})=\text{O})=\text{C}([\text{O-}])\text{C}1=\text{O}$, but it also has a pair of electrons delocalized on the five-membered ring and has certain aromaticity, so the stability is not extremely low, and option F is incorrect.

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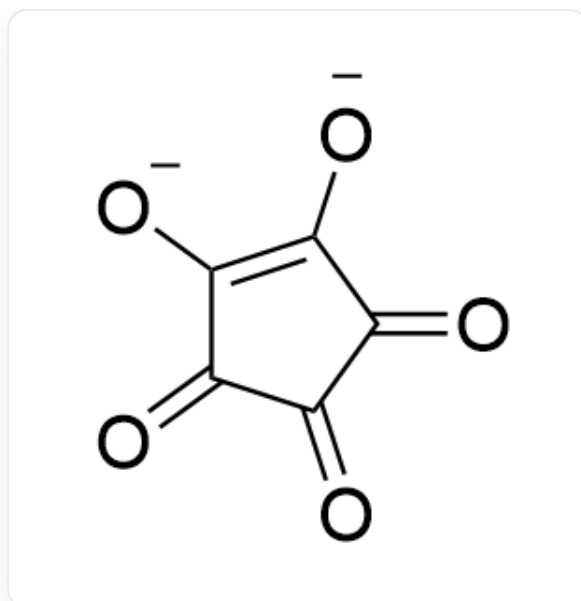
2 PTS

Y^{2-} is $\text{C}_5\text{O}_5^{2-}$, and the structure is $[\text{O-}]\text{C}(\text{C}(\text{C}1=\text{O})=\text{O})=\text{C}([\text{O-}])\text{C}1=\text{O}$

CHECKPOINT

1 PTS

Y^{2-} has a pair of electrons delocalized on the five-membered ring, and has certain aromaticity



The structure of $\mathbf{Y}^{2-}$: [O-]C(C(C1=O)=O)=C([O-])C1=O

A and $\text{Pb}_3(\text{OH})_2(\text{CO}_3)_2$ have the same constituent elements, so they both contain carbon, hydrogen, and oxygen elements as negative ions; **A** contains $\text{C}_6\text{O}_6^{2-}$, since it is a basic salt, considering the +2 valence of lead, and combining the stoichiometric ratio of each ion, the chemical formula of **A** can only be $\text{Pb}_2[\text{C}_6\text{O}_6](\text{OH})_2 \cdot \text{H}_2\text{O}$, which contains 18 atoms after removing the crystal water, so option B is incorrect.

CHECKPOINT

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The chemical formula of **A** is $\text{Pb}_2[\text{C}_6\text{O}_6](\text{OH})_2 \cdot \text{H}_2\text{O}$

In summary, options B-F are all incorrect, and option A is correct.